Windows Registry Change Monitoring System

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Abstract

This project implements a Windows Registry Change Monitoring System designed to detect unauthorized or suspicious modifications in critical registry locations. The system captures a baseline snapshot of sensitive registry keys and continuously monitors them using polling-based state comparison. It detects additions, deletions, and modifications in autorun and security-related registry paths. Detected changes are logged and alerted, simulating real-world endpoint monitoring mechanisms used in Blue Team cybersecurity operations.

Problem Statement

The Windows Registry stores critical system configuration data including startup programs, security policies, and firewall settings. Malware frequently modifies registry keys to achieve persistence, disable security tools, or manipulate system behavior. Without active monitoring, unauthorized registry changes may go unnoticed, increasing the risk of system compromise.

Objectives

- Monitor autorun registry keys for persistence detection.
- Detect malware-like configuration changes.
- Implement a baseline-based registry integrity checker.
- Generate alerts for suspicious modifications.
- Produce detailed change logs for forensic analysis.

System Architecture

The system consists of the following modules:

- Configuration Module
- Baseline Manager
- Monitoring Engine
- Change Detection Module
- Logging & Alerting Module

Workflow:

Load Config → Load Baseline → Monitor → Compare → Detect → Log → Repeat

Tools & Technologies Used

Programming Language: Python

Modules Used:

- winreg (Registry access)
- json (Configuration & baseline storage)
- hashlib (Integrity hashing)
- time (Polling interval control)

Implementation Details

Configuration File

Defines sensitive registry keys to monitor such as autorun and security-related paths.

Baseline Creation

Captures trusted registry snapshot and stores it in a JSON file for comparison.

Monitoring Engine

Reads registry values periodically and compares with baseline using polling mechanism.

Change Detection Logic

Identifies the following types of changes:

- ADDED values
- MODIFIED values
- DELETED values

Logging & Alerting

Writes detected changes to log files such as:

- logs/registry_changes.log
- logs/alerts.log

Practical Scope of the Project

- Monitor Run and RunOnce keys in HKCU and HKLM.
- Detect new startup entries added without authorization.
- Identify executables launched automatically upon login.
- Detect security policy modifications and firewall/Defender disable keys.
- Capture baseline registry snapshot and detect integrity violations.
- Generate detailed logs with timestamp, key path, old value, and new value.

Workflow Explanation

1. Define Sensitive Registry Keys
2. Create Baseline Snapshot
3. Start Continuous Monitoring

4. Detect Add/Modify/Delete Changes
5. Compare with Suspicious Patterns
6. Generate Alerts and Logs
7. Export Final Report

Expected Output

- Timestamped registry change logs
- Suspicious modification alerts
- Autorun persistence detections
- Integrity verification results

Learning Outcomes

- Understanding Windows Registry structure and key categories
- Knowledge of malware persistence techniques
- Registry-based attack detection methods
- Integrity monitoring techniques
- Blue Team defensive scripting skills

Conclusion

The Windows Registry Change Monitoring System successfully detects unauthorized changes in critical registry paths using a baseline-based integrity verification model. The system demonstrates practical endpoint monitoring techniques aligned with Blue Team cybersecurity practices.